

APÊNDICE A — MÉTRICAS DETALHADAS DO TREINAMENTO

Tabela 1 – Histórico de métricas de treinamento e validação por época

Época	Acurácia	Coef. Dice	IoU Score	Loss (Perda)	Val. Acurácia	Val. Coef. Dice	Val. IoU Score	Val. Loss	Tx. Aprend. (LR)
1	0.9549	0.9283	0.8753	0.0934	0.2968	0.4534	0.2968	73.9956	0.0010
2	0.9718	0.9504	0.9063	0.0618	0.3231	0.4656	0.3057	1.1038	0.0010
3	0.9708	0.9497	0.9047	0.0624	0.3252	0.4583	0.3013	0.6932	0.0010
4	0.9757	0.9584	0.9205	0.0488	0.9347	0.8983	0.8158	0.3381	0.0010
5	0.9727	0.9506	0.9068	0.0587	0.9241	0.8853	0.7952	0.2272	0.0010
6	0.9745	0.9562	0.9163	0.0520	0.9633	0.9426	0.8916	0.1847	0.0010
7	0.9749	0.9552	0.9154	0.0507	0.9551	0.9218	0.8552	0.1552	0.0010
8	0.9769	0.9596	0.9229	0.0462	0.9658	0.9373	0.8823	0.0950	0.0010
9	0.9762	0.9569	0.9184	0.0487	0.9660	0.9429	0.8923	0.0751	0.0010
10	0.9786	0.9610	0.9256	0.0448	0.9715	0.9464	0.8984	0.0897	0.0010
11	0.9819	0.9689	0.9400	0.0378	0.9810	0.9688	0.9396	0.0402	0.0010
12	0.9842	0.9724	0.9465	0.0337	0.9821	0.9639	0.9306	0.0401	0.0010
13	0.9848	0.9714	0.9457	0.0336	0.9766	0.9613	0.9258	0.0543	0.0010
14	0.9867	0.9764	0.9540	0.0281	0.9868	0.9775	0.9560	0.0292	0.0010
15	0.9871	0.9776	0.9563	0.0277	0.9787	0.9634	0.9294	0.0483	0.0010
16	0.9876	0.9794	0.9596	0.0254	0.9850	0.9742	0.9498	0.0309	0.0010
17	0.9883	0.9792	0.9593	0.0249	0.9847	0.9730	0.9475	0.0334	0.0010
18	0.9882	0.9792	0.9594	0.0253	0.9868	0.9756	0.9525	0.0288	0.0010
19	0.9879	0.9785	0.9580	0.0256	0.9821	0.9668	0.9360	0.0446	0.0010
20	0.9882	0.9790	0.9588	0.0251	0.9760	0.9613	0.9255	0.0708	0.0010
21	0.9880	0.9782	0.9576	0.0263	0.9821	0.9675	0.9372	0.0476	0.0010
22	0.9884	0.9807	0.9622	0.0239	0.9838	0.9707	0.9434	0.0399	0.0010
23	0.9865	0.9746	0.9509	0.0298	0.9750	0.9561	0.9161	0.0658	0.0010
24	0.9876	0.9783	0.9576	0.0263	0.9877	0.9796	0.9601	0.0251	0.0002
25	0.9891	0.9804	0.9616	0.0236	0.9855	0.9735	0.9486	0.0327	0.0002
26	0.9887	0.9810	0.9627	0.0233	0.9872	0.9779	0.9567	0.0277	0.0002

Época	Acurácia	Coef. Dice	IoU Score	Loss (Perda)	Val. Acurácia	Val. Coef. Dice	Val. IoU Score	Val. Loss	Tx. Aprend. (LR)
27	0.9890	0.9811	0.9629	0.0228	0.9864	0.9775	0.9561	0.0279	0.0002
28	0.9895	0.9809	0.9626	0.0226	0.9866	0.9762	0.9536	0.0302	0.0002
29	0.9889	0.9814	0.9634	0.0228	0.9842	0.9731	0.9477	0.0357	0.0002
30	0.9892	0.9812	0.9631	0.0225	0.9870	0.9764	0.9539	0.0284	0.0000
31	0.9897	0.9821	0.9649	0.0217	0.9859	0.9762	0.9536	0.0310	0.0000
32	0.9896	0.9822	0.9650	0.0219	0.9868	0.9757	0.9527	0.0301	0.0000
33	0.9897	0.9813	0.9634	0.0217	0.9870	0.9765	0.9543	0.0302	0.0000
34	0.9894	0.9824	0.9654	0.0217	0.9869	0.9787	0.9582	0.0280	0.0000
35	0.9893	0.9754	0.9561	0.0259	0.9872	0.9779	0.9568	0.0284	0.0000
36	0.9895	0.9822	0.9650	0.0217	0.9861	0.9774	0.9559	0.0300	0.0000
37	0.9898	0.9828	0.9663	0.0210	0.9872	0.9785	0.9579	0.0275	0.0000
38	0.9898	0.9813	0.9634	0.0219	0.9869	0.9765	0.9541	0.0311	0.0000
39	0.9897	0.9817	0.9642	0.0218	0.9869	0.9763	0.9539	0.0299	0.0000

Fonte: Dados do treinamento (2026).